

MOHAMMED FAYIZ PARAPPAN

1st year PhD Student, Electrical & Computer Engineering · Duke University

[Email](#) · [Google Scholar](#)

RESEARCH INTERESTS

Pluralistic AI Alignment · Subjective NLP · Human-AI Interaction · Value Calibration · Sociocultural Fairness in ML

EDUCATION

Duke University

Aug 2025 – Present

PhD, Electrical & Computer Engineering | GPA: 4.0/4.0 | Advisor: Dr. Ricardo Henao

Coursework: Deep Learning (ECE 685D), Natural Language Processing (ECE 684), Transformers to LLMs (ECE 590)

IIT Kharagpur

Nov 2020 – May 2025

Integrated BTech & MTech, Industrial & Systems Engineering | CGPA: 9.46/10 (Department Rank 1)

Courses: Multivariate Stat. Modelling, Stat. Learning, Recommender System, Safety Analytics, Generalized Linear models

PUBLICATIONS

Labels have Human Values: Value Calibration of Subjective Tasks

Mohammed Fayiz Parappan & Ricardo Henao | Under Review

- Proposed MC-STL, a framework that identifies latent human value groups in subjective NLP annotations (via label rationale clustering, expert value taxonomies, or sociocultural descriptors) and enforces value-conditional calibration across binary, ordinal, and preference learning tasks.
- Demonstrates significant improvements in predictive performance, multicalibration over diverse value groups and stronger minority-opinion alignment on subjective NLP benchmarks covering toxic chatbot conversations, value reasoning, Text-to-Image (T2I) safety, and RLHF preference pairs.
- Directly addresses risks of NLP categorical decision systems (that do not generate text) suppressing minority but legitimate human values by preserving latent value diversity during model training.

Learning Subjective Label Distributions via Sociocultural Descriptors

Mohammed Fayiz Parappan & Ricardo Henao | [EMNLP 2025 Main \(CORE A*, Poster\)](#)

- Proposed LSLD, a novel framework for modeling subjectivity in binary labeling task from a text item as an empirical probability distribution, incorporating language model generated value perspectives behind text item and annotator's sociocultural backgrounds.
- Comprehensive evaluation against existing baselines demonstrates LSLD's improved individual probabilistic prediction, well calibrated toxicity distribution prediction on text items.
- The framework preserves overall population level and sociocultural subgroup level viewpoints of toxicity on text items.

RESEARCH EXPERIENCE

Visiting Researcher, KAUST AI Initiative

2024

Master Thesis Collaboration

- Developed Toxicity Distribution Modelling approach, addressing aggregation biases by training on distribution of diverse labels for each text item via inclusion of sociocultural backgrounds in training setup.

Data Science Fellow, ONGC (Industry Deployment)

2024

- Built deep learning pipeline (U-Net with focal loss) for seismic fault segmentation deployed in industry; applied instance-weighting transfer learning across domain-shifted datasets.

MITACS Globalink Fellow, Queen's University, Ontario

May 2023 – Feb 2024

Few-Shot Meta-Learning for EEG/BCI Classification

- Implemented Siamese Networks and Model Agnostic MetaLearning for arrhythmia classification (52 subjects, 4 classes); achieved 98.8%–99.5% accuracy on MIT-BIH and Incart datasets.
- Built Dempster-Shafer Theory fusion for multi-view uncertainty reasoning; performed ablation studies on decision fusion algorithms under Gaussian noise conditions.

Research Assistant, Brain-Computer Interface Lab, IIT Kharagpur

Mar 2022 – Dec 2023

- Compared EEGNet, CTCNN, ShallowFBCSPNet, Deep4Net on within-subject and cross-subject Motor Imagery classification paradigms.
- Built MAML framework over EEGNet for low-resource new-subject adaptation in few-shot settings.

ADDITIONAL PUBLICATIONS

- Unveiling Subsurface Faults in Indian Krishna Godavari Basin via Domain Adaptation — Ghosh T., Mohammed Fayiz et al. · IEEE TGRS (Q1, IF: 8.5)
- Uncertainty Aware Multi-View Arrhythmia Classification from ECG — Ashhad M., Mohammed Fayiz, Etemad A., Hashemi J. · IJCNN 2024 (ERA Rank: A)
- Deep Learning Pipeline for EEG Classification: Evaluating Models and Subject Generalizability — Mewade D., Mohammed Fayiz, Samanta D. · ICPR 2024 (ERA Rank: A)
- Instance Weighting Knowledge Transfer Network for Seismic Fault Detection · NeurIPS Affinity Workshop 2024

TECHNICAL SKILLS

Languages & Frameworks: Python (PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, HuggingFace Transformers, LangChain, OpenCV), R (Statistical Modelling), Matlab

AWARDS & FELLOWSHIPS

Duke PhD Fellowship, Duke University	2025
MeitY IndiaAI Fellow, Ministry of Electronics & IT, India	2025
Director's Institute Silver Medal (Dept. Rank 1), IIT Kharagpur	2025
KAUST AI Visiting Researcher Award	2024
MITACS Globalink Award, Queen's University, Ontario	2023
Ranked 19th, Amazon India ML Competition (70,000+ participants)	2024